

Syllabus

Week 1 – Visualizing Data Using Spreadsheets

- Course Intro
- Optional Lab: Excel Basics
- Reading: Excel Keyboard Shortcuts
- Introduction to Charts
- Interview: Using Visualizations to Tell a Story
- Creating Basic Charts in Excel
- Using the Excel PivotChart Feature
- Lab: Creating Basic Charts
- Practice Quiz: Visualizing Data using Spreadsheets
- Graded Quiz: Visualizing Data using Spreadsheets

Week 2 – Creating Visualizations and Dashboards with Spreadsheets

- Creating Treemaps, Scatter Charts, and Histograms
- Creating Filled Map Charts and Sparklines
- Lab: Creating Advanced Charts
- Practice Quiz: Creating Advanced Charts
- Graded Quiz: Updating Advanced Charts
- Introduction to Dashboarding
- Interview: Using Dashboards to Present Data Results
- Creating a Simple Dashboard using Excel
- Lab: Creating a Simple Dashboard with Excel

- Practice Quiz: Creating Dashboards Using Spreadsheets
- Graded Quiz: Creating Dashboards Using Spreadsheets

Week 3 – Creating Visualizations and Dashboards with Cognos Analytics

- Cognos Analytics: Introduction and How to Sign Up
- Navigating in Cognos Analytics
- Lab: Getting Started with Cognos Analytics
- Practice Quiz: Getting Started with IBM Cognos Analytics
- Creating a Simple Dashboard in Cognos Analytics
- Lab: Different Methods for Creating Dashboard Visualizations with Cognos Analytics
- Advanced Capabilities in Cognos Analytics Dashboards
- Lab: Advanced Dashboard Capabilities in Cognos Analytics
- Practice Quiz: Creating Dashboards with Cognos Analytics
- Graded Quiz: Creating Visualizations and Dashboards with Cognos Analytics

OPTIONAL CONTENT

- Reading: About this optional lesson with Looker Studio
- Reading: Getting Started with Google Looker Studio
- Lab: Getting Started with Google Looker Studio
- Creating Visualizations in Reports using Looker Studio
- Lab: Creating and Configuring Visualizations with Google Looker Studio
- Practice Quiz: Creating and Configuring Visualizations with Google Looker Studio

Week 4 – Final Project - Creating Visualizations and Dashboards

- Reading: Final Project Overview
- Final Assignment – Part 1: Creating Visualizations Using Excel
- Final Assignment – Part 2a: Creating Visualizations Using Cognos Analytics OR
- Final Assignment – Part 2b: Creating Visualizations Using Google Looker Studio
- Final Assignment – Part 3: Peer-graded Assignment – Learner Submission
- Final Assignment – Part 3: Peer-graded Assignment – Peer Review

The table below lists keyboard shortcuts for some of the most common Excel tasks.

Task	Shortcut
Close a workbook	Ctrl+W
Open a workbook	Ctrl+O
Save a workbook	Ctrl+S
Copy	Ctrl+C
Cut	Ctrl+X
Paste	Ctrl+V
Undo	Ctrl+Z
Remove cell contents	Delete
Bold	Ctrl+B
Open context menu	Shift+F10
Expand or collapse the ribbon	Ctrl+F1

Move up one cell in the worksheet	Up arrow key
Move down one cell in the worksheet	Down arrow key
Move one cell left in the worksheet	Left arrow key
Move one cell right in the worksheet	Right arrow key
Move to the edge of the current data region in the worksheet (e.g. end of column)	Ctrl+Arrow key (e.g. Ctrl+Down arrow)
Move to the last cell on a worksheet	Ctrl+End
Move to the beginning of a worksheet	Ctrl+Home
Extend the selection of cells to the last used cell on a worksheet (lower right corner)	Ctrl+Shift+End
Move to the cell in the upper-left corner of the window (when Scroll Lock is On)	Home+Scroll Lock
Move one screen down in a worksheet	Page Down
Move one screen up in a worksheet	Page Up
Move one screen to the right in a worksheet	Alt+Page Down
Move one screen to the left in a worksheet	Alt+Page Up
Move to the next sheet in a workbook	Ctrl+Page Down
Move to the previous sheet in a workbook	Ctrl+Page Up

Edit the active cell and put the cursor at the end of the cell's contents	F2
Enter the current time	Ctrl+Shift+colon (:)
Enter the current date	Ctrl+semi-colon (;)

Introduction to charts

- I. Line charts
 - A. Compare different but related data sets
 - B. Display trends
 - C. Show how data values change in relation to continuous variable (time)
 - D. Show trends in data at equal time intervals
 - E. Data that's arranged in columns or rows.
- II. Pie charts
 - A. Can show breakdown of entity into sub-parts
 - B. Proportion of sub parts in relation to each other
 - C. Each portin represents static value or category
 - D. Sum of all categories equals 100%
 - E. Example: marketing
 1. Social sites
 2. Native advertising
 3. Paid influencers
 4. Live events
- III. Bar charts
 - A. Most common
 - B. Easy to creat
 - C. Compare related data sets or parts of whole
- IV. Stacked bar charts
 - A. Each bar divided into sub-bars
 - B. Stacked end to end
- V. Column charts
 - A. Show change over time
 - B. Compare values side-by-side
 - C. Similar to bar, but not always interchangeable
 - D. Shows negative and positive values, use bar
- VI. Treemaps
 - A. Useful for displaying complex hierarchies using nested rectangles
- VII. Funnel charts
 - A. Pipeline or different stages of a continuous process

- VIII. Scatter charts
 - A. Colors represent categories of data
 - B. Circle sizes represent volume of data
 - C. Reveal trends, clusters, patterns, and correlations
- IX. Bubble charts
 - A. Variant of scatter chart
 - B. Useful for comparing handful of categories to one another
- X. Sparklines
 - A. Don't include axis
 - B. Display trends simply and effectively
 - C. General trend of a variation

Creating basic charts in Excel

- I. Line graphs
 - A. Select two different columns you want as line graph.
 - B. Go to insert>line chart
 - C. Can change title
- II. Pie Chart
 - A. Simple visualization of differing data results
 - B. Best used when you only have one data series
 - C. Data contains no more than a dozen data categories
 - D. To make
 - 1. Select columns
 - 2. Select pie chart from 2-D pie
 - 3. Change chart design style
- III. Bar chart
 - A. Categories arranged on vertical axis, values on horizontal
 - 1. Columns is opposite
 - B. Select style of 2-d bar chart to display
 - 1. Can change color and titles
 - 2. Can change color and choose a color pallet

Using Excel PivotChart Feature

- I. Pivot charts show data series, categories, and chart axes just like standard charts do
 - A. Source data is hosted in pivot table
 - B. Pivot charts are useful for making sense of complex data in pivot table
- II. Area chart
 - A. Info
 - 1. Show info as series of data points connected by straight lines
 - 2. Have a filled area below the connected data points
 - 3. Can show both positive and negative values
 - B. How to
 - 1. First create copy of pivot
 - 2. Filter data for needed info

3. Create area chart
4. PivotTable Analyze>PivotChart> insert chart>3-D area chart
 - a) Floating chart showing area type
 - b) Can filter data in pivot chart rather than in pivot table
 - c) In pivot chart can filter to only display one type of data
5. Column chart
 - a) Compares values across categories using vertical bars
 - (1) Categories are typically on horizontal axis
 - (2) Values are typically on vertical axis
 - b) To do
 - (1) Select 3-D column chart
 - (2) Can collapse or expand all values, not individual ones, within chart

Summary and Highlights

In this lesson you have learned:

- The importance of charts and how they are able to shape our data.
- How to use visualizations to provide meaningful information.
- How to use Excel to create line, pie, and bar charts.
- About Excel pivot tables in creating area and column charts using the PivotChart feature.
- How to use a pivot table or pivot chart to filter data and how to expand and collapse data levels.

Creating Treemaps, scatter charts and histograms

I. Treemaps

A. Definition

1. Compare values across hierarchy levels
2. Show proportions tables
3. Good way to display lots of data in one graphic
4. Color, size, and closeness of shapes represent hierarchical data categories

B. To create a treemap

1. Select two columns
2. Select treemap chart from hierarchy charts from charts group
3. Change chart design withing ribbon

II. Scatter chart

A. Definition

1. Compare two sets of numerical data values
2. Shows relationships between them
3. Combines two sets of x and y axis values into single data points
4. Displayed in clusters
5. Commonly used to compare statistical, scientific, and engineering data

B. To create

1. Select scatter from chart type
2. Change chart style
3. Can add x and y axis title under add chart element

III. Histogram

A. Define

1. Show distribution of data grouped into "bins"
2. Looks similar to bar chart, but are very different
3. Bar chart compares data, histograms displays distribution of data

B. To create

1. Select 2 columns
2. Select histogram from statistical section of charts
3. Open format pane, in axis options, can change the bin width or number of bins

Creating filled map charts and sparklines

I. Filled map charts

- A. Used to compare values and show categories across geographical regions
 - 1. Countries, states, postal codes
 - B. Copy-paste data from pivot table
 - 1. Charts group> filled map chart
 - 2. Can change colors on it
- II. Sparklines
 - A. Are mini charts showing in single cells
 - 1. Used for showing trends
 - a) Price fluctuations,
 - b) Can show max and min values
 - c) Works best when put next to data it represents
 - B. Select line type from group
 - 1. Specify where want it to show up
 - 2. Can copy-paste down and will use updated data
 - C. Maximum and minimum values to show on spark lines
 - 1. Several styles
 - 2. Under sparkline, check high and low point
- III. Other charts
 - A. Waterfall chart
 - 1. Cumulative effect of series of positive and negative data
 - 2. Show inflow and outflow like financial data
 - B. Funnel type
 - 1. Show progressively smaller stages in a process, progressively reducing data
 - C. Stock price
 - 1. Show trend of stocks performance over time
 - 2. Data with multiple stock price values like open, close, volume, high, and low
 - D. Surface
 - 1. Used to show trends in values in 2-d contour charts
 - 2. When categories and data series are both numeric
 - E. Radar
 - 1. Show data relative to center point
 - 2. Suitable when categories are not directly comparable

Summary and Highlights

In this lesson you have learned:

- How to create treemaps, scatter charts, and histograms in Excel.
- About the different charts available in Excel, and how each one is different in displaying data.
- How to create filled map charts and sparklines in Excel.

Introductions to dashboards

I. Dashboard

A. from car manufacturers

1. Used easy to understand visuals to understand what's going on

B. Data analysis dashboards provide key information in one place

1. Can provide interaction capabilities (tools on dashboard)
2. Get consolidated and visualized insight into their most important business data and KPIs key performance indicators
3. Are controllable self-service business intelligence BI interface through use of filters
 - a) Typically created in data analysis application by using multiple pivot tables and charts
 - b) Could be used from one data source or multiple data source

C. Benefits

1. Offer insights into data
2. Can alert you to trends and patterns
3. Provide interactive experience
4. Dynamically updated as data changes
5. Provide centralized consolidated view of business data

D. Dashboarding skill is a "must have"

E. Basic spreadsheet apps

1. Excel, google sheets
2. Advanced data analysis and visualization
 - a) Bokeh, Dash in python, R-studio's shiny, Tableau, etc.

Creating a simple dashboard using excel

I. Setting up a dashboard

A. First

1. First collect and organize data
2. Verify data is clean, error-free, has no blank rows or columns
3. Format as data table
4. Create pivot tables
 - a) Highlight key aspects of data
5. Create several visualizations to populate dashboard

B. To make

1. Make a new tab called dashboard
 - a) Copy-paste visuals from other sheets to new tab
 - b) Make them look better by resizing objects
 - (1) "Less is more"
 - c) Make a copy of the dashboard
 - (1) Thin down
 - (2) Make visualizations more focused
 - (3) Delete a few superfluous chart that have similar data or other data that are not key to the message
 - (4) Copy-paste slicer from chart that is in the dashboard so we can have some interactivity
 - (5) Resize and change colors of visuals by applying same style to all elements, can even recolor slicer
 - (6) Remove excel interface. Can remove gridline, formula bar, and headings
 - (7) When using slicer and changing data, can see that any data that is linked to it will be updated. Will also change in other tabs as well

Cognos Analytics: Introduction and how to sign up

I. Define

- A. Business intelligence and analytics platform
- B. Offers range of tools
 - 1. Data modeling
 - 2. Data exploration
 - 3. Advanced analytics
 - 4. Natural language generation
 - 5. Interactive dashboards and custom reporting

Navigating in Cognos Analytics

I. Two navigation areas

- A. Main menu
- B. Toolbar at top right of page
 - 1. Check how many days remain in trial
 - 2. AI assistant

II. Create content

- A. Upload data
- B. Create new content

III. Main menu in top left

- A. New dashboard or files
- B. Upload data
- C. Content

IV. To upload a file

- A. Left menu
- B. Upload data
- C. Select to file >open
- D. Uploaded content is stored in My content under menu

V. Content

- A. Team content shows any shared content

VI. Can use assistant to look at data once it's uploaded

VII. Creating a dashboard

- A. Select dataset
- B. From menu>create>dashboard
- C. Select a template
- D. Left pane shows data fields
 - 1. "Selected source"

2. Sources icon reviews and shows selected data
3. Pins show visualizations you've pinned
4. Visualizations
 - a) System tab shows visualizations can use
 - b) Can upload custom one as well to other tab
5. To pin an item
 - a) Add visualization
 - b) Select pin icon in ribbon menu
6. Widgets
 - a) Can add widgets to dashboard and stories
 - (1) Text, visions and shapes, audio media, etc.

Creating a simple dashboard in cognos analytics

- I. Select dataset
 - A. In toolbar choose to create a dashboard
 - B. Select relevant dashboard
 - C. Click create
 1. Usage property for each data content can be changed
 - a) Go to 3 dot bar to change
 - D. Choose a data item and drag onto the canvas
 1. When dragging item, need to make sure it turns blue before dropping
 2. Can see if other recommended visualizations, but want to turn it into summary, can select from top menu ribbon
 3. Can change from sum to average, etc
 - E. Can select specific visualization by choosing from graph/bar graph icon
 - F. For example by geography
 1. Can drag sales into the content panel
 2. Can configure properties in pane on right
- II. Can also use assistant to make visualizations
 - A. Will offer up insights we haven't thought about
 - B. Can ask a question and it will show you data visualizations
 - C. Can drag visualization onto canvas
- III. All visualizations are meant to be auto interactive/updated
 - A. Can select something from one chart and it will auto-update all other graphics

- B. Can also add filters (upper right ribbon)
- C. Can drag something from the list of data and drag into the tab to

Advanced capabilities in Cognos analytics dashboards

- I. Select dataset and make dashboard
- II. Change usage property of data if needed
- III. Can do calculation
 - A. Right click datafile and select calculation
 - B. Give a name
 - C. Functions list our different options
 - D. Can type out and will suggest or drag and drop
 - E. Drag and drop from left pane or type out and select from suggestion list.
 - F. Will be added to data available
 - G. Can drag and drop
- IV. Can create a navigation path to drill up and down on data
 - A. Right click and add new data path
 - B. Select navigation path, right click on “trains” and select Drill down
 - C. Can right click on axis and select drill up
- V. Can exclude item by right clicking and select exclude in legend
- VI. Top or bottom filters
 - A. Focus only on most crucial data points
 - B. TO use
 - 1. Right click on x or y axis
 - 2. Select top or bottom
 - 3. Select top and will show top
- VII. Instantly generate infographics
 - A. Choose widget from left to make it a different infographic

Summary and Highlights

In this lesson you have learned:

- How to use Cognos Analytics to create a simple dashboard.
- About the value of Cognos Analytics and its advanced capabilities, such as creating calculations and leveraging navigation paths.

About this optional lesson with Looker Studio

In the previous lesson you became familiar with Cognos Analytics to create visualizations and dashboards. Depending on your project and employer requirements you may need to use other BI tools for the job such as Tableau, Microsoft Power BI, Google Looker Studio, Zoho Analytics etc.

To help you develop additional experience in this optional lesson you will work with Google Looker Studio which you can get started with at no charge.

Although it is not required for you to complete this lesson to successfully complete the course, there is an option in the final project to use either Cognos Analytics or Google Looker Studio. Therefore, if you prefer to complete the final project using Looker we recommend that you complete this lesson and attempt all the labs in it.

Summary and Highlights

In this lesson you have learned:

- The fundamentals of the Looker Studio application.
- How to navigate around the Looker Studio user interface.
- How to create reports in Looker Studio.

- How to connect to data sources in Looker Studio.
- How to create dashboards in Looker Studio.